simeon djambov

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research

❖ **postdoctoral associate** 2026–
Laboratoire d'Hydrodynamique de l'Ecole Polytechnique (LadHyX) Palaiseau, France

· advisor: Etienne Jambon-Puillet;

❖ **doctoral assistant** 2021–2025
Laboratory of Fluid Mechanics and Instabilities, EPFL Lausanne, Switzerland

· thesis title: "Gravity-driven thin films: flow and dissolution;"
· director: François Gallaire;

❖ **visiting researcher** Feb–Aug 2024
Laboratoire Matière et Systèmes Complexes Paris, France

· host: Sylvain Courrech du Pont;

❖ **research intern** 2017–2021

· "Fingering instability on a sphere" (F. Gallaire, LFMI, Feb–Jul 2021);
· "Impact-induced cavitation" (C. Raufaste & F. Celestini, INPHYNI, Apr–Jul 2020);
· "Electric double-layer dynamics" (A. Würger, LOMA, May–Jun 2019);
· "Turbulent friction in 2D rough-walled flows" (H. Kellay, LOMA, May 2018–Apr 2019);
· "Femtosecond laser spectroscopy" (J. Degert, LOMA, May–Jun 2017);

education

❖ **ph.d. in mechanics** 2021–2025
EPFL Lausanne, Switzerland

❖ **m.sc. in physics** 2019–2021
Université Côte d'Azur Nice, France

❖ **b.sc. in physics** 2016–2019
Université de Bordeaux Bordeaux, France

❖ **highschool diploma** 2011–2016
French language Highschool "Alphonse de Lamartine" Sofia, Bulgaria

languages

bulgarian (native); french (fluent); english (proficient); german (limited);

interests

piano (ABRSM, Grade 8), choir (CHUL), F. M. Dostoevsky, pétanque, sudoku, sailing, swimming;

publications

[5] S. Djambov, A. Marcotte, F. Gallaire, P. G. Ledda. "Thin-film drainage becomes singular at saddles," *Physical Review Letters* **137**, 164002, 2026.

[4] S. Djambov, F. Gallaire. "Analytical solutions to the gravity-driven rain-fed free-surface Stokes equations," *Comptes Rendus. Mécanique* **354**, 451–459, 2026.

[3] S. Eghbali, **S. Djambov**, F. Gallaire. "Stability of a liquid layer draining around a horizontal cylinder: Interplay of capillary and gravity forces," *Physical Review Fluids* **9**(6), 063903, 2024.

[2] S. Djambov, F. Gallaire. "Boundary-layer approach to the linear karren instability," *Journal of Fluid Mechanics* **977**, R3, 2023.

[1] A. Vilquin, J. Jagielka, **S. Djambov**, H. Herouard, P. Fischer, C.-H. Bruneau, P. Chakraborty, G. Gioia, H. Kellay. "Asymptotic turbulent friction in 2D rough-walled flows," *Science Advances* **7**(5), eabc6234, 2021.

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[0] M. Chaigne, A. Maller, **S. Djambov**, M. Berhanu. "Dissolution rate of a soluble surface in a unidirectional turbulent flow," *under review in Journal of Fluid Mechanics*.

[0] S. Djambov, F. Gallaire. "Linear instability of weakly non-parallel film flows," *under review in Journal of Fluid Mechanics*.

[0] S. Djambov, L. Hasbroucq, F. Gallaire, S. Courrech du Pont. "Shape evolution of soluble blocks under rainfall," *in preparation*.

teaching

❖ **epfl:** ME-444 Hydrodynamics (F. Gallaire; 2023, 2025); ME-466 Instability (F. Gallaire; 2023, 2024); ME-446 Two-Phase Flows and Heat Transfer (F. Gallaire; 2022); PHYS-106(d) Thermodynamics (S. Bréchet; 2022); PHYS-101(d) Mechanics (P. Müllhaupt; 2021);
co-supervision of two master's students (2023, 2024);

grants

❖ **epfl doc.mobility:** "Condensation on soluble substrates: a path to impact-less linear karren formation;" six-month-long stay funded for nearly 20 kCHF;